

Reflective Journal

Joseph Clay - ITAI 1371 - Lab 05

What prep step changed the data most?

Imputing Age (177 of 891 missing, ~20%) and one-hot encoding Sex/Embarked. Scaling changed values but not information.

Where did silent failure risk show up?

- The handout's `df['Age'].fillna(median, inplace=True)` left all 177 NaN on pandas 3.0.6 with only a warning. I switched to explicit assignment and added an assert (ADR 0001).
- 2 missing Embarked values would have encoded as all zeros, silently labelled Cherbourg. I filled with the mode first (ADR 0002).
- Scaler fit on all rows = leakage if a train/test split existed.

How this prepares me for the Mid Term EDA (train only, test untouched)?

Impute, encode and scale must be fit on train only and then applied to test; I would use `SimpleImputer` and a pipeline there.

Process: kept my spec/gate/RCA/ADR workflow; DoD came from Canvas, not Module 3.